

Exact Continuous Majorization on the Simple-Spectrum Stratum

Candidate partial result for arXiv:2210.13309

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Status. Candidate partial result, likely valid, pending human review. Question 7.2 of the source has an affirmative answer whenever the majorizing field has simple spectrum at every point. In that case the desired element is the average of exactly n continuous unitary conjugates, with all weights equal to $1/n$. The argument does not treat eigenvalue collisions.

1 Source question

Mootoo and Skoufranis study several versions of joint majorization in $C(X, M_n(\mathbb{C}))$ and show that pointwise majorization characterizes the *closed* convex hull of a joint unitary orbit [1]. Their Question 7.2 asks whether closure can be removed for one self-adjoint operator:

Let $A, B \in C(X, M_n)$ be self-adjoint and continuously diagonalizable. Does $A \prec_c B$ imply that $A \in \text{conv}(\mathcal{U}(B))$?

$C(\mathcal{X}, \mathcal{M}_n)$ are always CSD if and only if $\mathcal{X}$ is sub-Stonean with $\dim(\mathcal{X}) \leq 2$ and carries no non-trivial G -bundles over any closed subset for G a symmetric group or the circle group. It would be quite interesting if the answer to Question 7.1 was the same condition.

Although Corollaries 3.4 and 3.5 demonstrate that $\vec{A} \prec_c \vec{B}$ does not $\vec{A} \in \text{conv}(\mathcal{U}(\vec{B}))$ when $\vec{A}$ and $\vec{B}$ have length (greater than or equal to) 2, we were unable to find an example for single operator majorization. In particular, the following question is still open:

Question 7.2. *Let $A, B \in C(\mathcal{X}, \mathcal{M}_n)$ be self-adjoint and continuously diagonalizable. Does $A \prec_c B$ imply that $A \in \text{conv}(\mathcal{U}(B))$?*

Of course there are a number of barriers to overcome for an affirmative answer to Question 7.2: unitary orbits in infinite dimensional C^* -algebras are often not closed; the operators here need not be continuously diagonalizable unless $\mathcal{X}$ is sub-Stonean etc.; the number of unitaries used and probability vectors need not be well-behaved. In terms of the last condition, note the technique developed in Lemma 7.3 required finiteness and issues with the infinite number of probability vectors one needs to consider was the core of the issue in Corollary 3.4. However, as the following two results show, some of these might not be issues.

Figure 1: Question 7.2 in the published source, page 20.

Here $A \prec_c B$ means that, after continuous diagonalizations, a continuous doubly stochastic matrix field sends the eigenvalue vector of B to that of A . The conclusion $A \in \text{conv}(\mathcal{U}(B))$ requires one finite probability vector, independent of $x \in X$, and continuous unitary fields.

2 Result

Theorem 1 (Simple-spectrum affirmative answer). *Let X be a compact Hausdorff space, and let $A, B \in C(X, M_n(\mathbb{C}))$ be self-adjoint and continuously diagonalizable. Assume that $A \prec_c B$ and that $B(x)$ has simple spectrum for every $x \in X$. Then there exist unitaries $U_0, \dots, U_{n-1} \in C(X, M_n(\mathbb{C}))$ such that*

$$A = \frac{1}{n} \sum_{k=0}^{n-1} U_k^* B U_k.$$

In particular, $A \in \text{conv}(\mathcal{U}(B))$, with n equal weights.

The conclusion is stronger than merely showing that A belongs to an unspecified finite convex hull. It provides a number of summands depending only on the matrix size and a fixed probability vector.

3 Idea of the proof

For a strictly ordered vector $b = (b_1, \dots, b_n)$, the possible diagonals of the Hermitian orbit of $D_b = \text{diag}(b)$ form the permutahedron

$$P(b) = \text{conv}\{\sigma b : \sigma \in S_n\}.$$

The Schur–Horn theorem gives surjectivity of the diagonal map onto $P(b)$, but a pointwise choice of a conjugating unitary is not enough here: it must vary continuously with both a and b .

The key observation is that a single compact slice works simultaneously for all strictly ordered b . Take the closure of a generic diagonal-torus orbit in the full flag variety. This is the permutohedral toric variety, and its nonnegative part maps homeomorphically to $P(b)$ under the b -weighted moment map. That nonnegative part is contractible, so its flag bundle admits a continuous unitary lift. We thereby obtain a jointly continuous unitary $R(a, b)$ for which

$$\text{diag}(R(a, b) D_b R(a, b)^*) = a.$$

Finally, averaging this matrix over the n cyclic diagonal phase matrices kills every off-diagonal entry without changing its diagonal. Each summand remains a unitary conjugate of D_b .

4 A uniform toric Schur–Horn slice

Let

$$\mathcal{C} = \{b \in \mathbb{R}^n : b_1 > b_2 > \dots > b_n\}.$$

Lemma 1 (Uniform slice). *There are a compact contractible space T_n and a continuous map $R : T_n \rightarrow U(n)$ such that, for every $b \in \mathcal{C}$, the map*

$$\theta_b : T_n \longrightarrow P(b), \quad \theta_b(t) = \text{diag}(R(t) D_b R(t)^*),$$

is a homeomorphism. Moreover, for every compact $K \subset \mathcal{C}$, the inverse

$$\{(b, a) : b \in K, a \in P(b)\} \longrightarrow T_n, \quad (b, a) \longmapsto \theta_b^{-1}(a)$$

is continuous.

Proof. Let $\text{Fl}(n) = U(n)/\mathbb{T}^n$ be the full flag variety, and let Z be the closure of a generic orbit of the complex diagonal torus. This is the permutohedral projective toric variety. Its moment image for a strictly dominant weight b is the weighted permutohedron $P(b)$; the generic-orbit closure and weighted-polytope description are recalled explicitly in [2, Sections 2.3–2.4].

Let $T_n = Z_{\geq 0}$ be the nonnegative part of this toric variety. The standard toric moment-map theorem says that the moment map restricts to a homeomorphism from $Z_{\geq 0}$ onto its moment polytope; see [3, Theorem 8.5]. Under the usual identification of the coadjoint orbit through D_b with $\text{Fl}(n)$, this moment map is precisely

$$[Q] \mapsto \text{diag}(QD_bQ^*).$$

Consequently the restriction is a homeomorphism $T_n \rightarrow P(b)$. In particular, T_n is homeomorphic to a polytope and hence contractible.

The quotient map $U(n) \rightarrow \text{Fl}(n)$ is a principal $\mathbb{T}^n$ -bundle. Its restriction to the contractible compact space T_n is trivial, so the inclusion $T_n \hookrightarrow \text{Fl}(n)$ has a continuous lift $R : T_n \rightarrow U(n)$. With this lift the moment map has the displayed form θ_b .

It remains only to record joint continuity of the inverse. For compact $K \subset \mathcal{C}$, consider

$$\Theta : K \times T_n \longrightarrow \mathbb{R}^n \times \mathbb{R}^n, \quad \Theta(b, t) = (b, \theta_b(t)).$$

This map is continuous and, because every θ_b is a homeomorphism, is a bijection onto

$$E_K = \{(b, a) : b \in K, a \in P(b)\}.$$

The domain is compact and E_K is Hausdorff. Hence Θ is a homeomorphism onto E_K , proving the final assertion. $\square$

5 Proof of Theorem 1

Choose continuous diagonalizations

$$A = V^*D_aV, \quad B = W^*D_bW,$$

where $a, b : X \rightarrow \mathbb{R}^n$ and $V, W : X \rightarrow U(n)$ are continuous. Since the spectrum of $B(x)$ is simple, the ordering of the coordinates of $b(x)$ is locally constant. Thus X is the disjoint union of at most $n!$ clopen sets on each of which a fixed permutation makes

$$b_1(x) > \cdots > b_n(x).$$

It suffices to work on each clopen piece and paste the resulting continuous unitaries, so assume this order globally.

The relation $A \prec_c B$ provides a doubly stochastic matrix $D(x)$ with $a(x) = D(x)b(x)$. By the finite-dimensional majorization theorem, $a(x) \in P(b(x))$. Put $K = b(X)$. Compactness of X and simplicity of the spectrum imply that K is a compact subset of $\mathcal{C}$. Lemma 1 therefore gives a continuous map

$$t(x) = \theta_{b(x)}^{-1}(a(x)), \quad Q(x) = R(t(x)),$$

such that

$$\text{diag}(Q(x)D_{b(x)}Q(x)^*) = a(x). \tag{1}$$

Let $\omega = e^{2\pi i/n}$ and

$$\Omega = \text{diag}(1, \omega, \omega^2, \dots, \omega^{n-1}).$$

For every matrix $H \in M_n(\mathbb{C})$, the finite Fourier identity gives

$$\frac{1}{n} \sum_{k=0}^{n-1} \Omega^{-k} H \Omega^k = \text{diag}(H). \quad (2)$$

Indeed, the (r, s) entry is multiplied by $n^{-1} \sum_k \omega^{k(s-r)}$, which is 1 for $r = s$ and 0 otherwise. Apply (2) to $H = QD_bQ^*$ and use (1):

$$D_a = \frac{1}{n} \sum_{k=0}^{n-1} \Omega^{-k} QD_bQ^* \Omega^k.$$

Define

$$U_k = W^* Q^* \Omega^k V \quad (0 \leq k < n).$$

These are continuous unitary fields, and

$$\frac{1}{n} \sum_{k=0}^{n-1} U_k^* B U_k = V^* \left(\frac{1}{n} \sum_{k=0}^{n-1} \Omega^{-k} QD_bQ^* \Omega^k \right) V = V^* D_a V = A.$$

The constructions on the finitely many clopen order pieces paste without any boundary condition, completing the proof.

6 Scope and remaining boundary

The proof only uses the pointwise consequence $a(x) \in P(b(x))$ of $A \prec_c B$; it does not otherwise use the particular continuous doubly stochastic witness. It therefore also proves the same conclusion under any hypothesis that supplies pointwise majorization together with the stated continuous diagonalizations and simple spectrum of B .

The unresolved case is precisely where the eigenvalue vector $b(x)$ reaches a wall of the Weyl chamber. At such a collision the weighted permutohedron changes combinatorial type and the toric moment map collapses faces, so the compact-to-Hausdorff inverse argument in Lemma 1 no longer applies. This is a real limitation of the proof, not merely omitted notation. The source already settles the $n = 2$ case without a simplicity assumption; the theorem above gives all dimensions on the generic no-collision stratum.

7 Verification and literature boundary

The script `code/verify_dephasing.py` checks the Fourier dephasing identity and the final conjugation formula for random unitary data in sizes 2 through 8. These checks validate the algebra after the toric slice has been selected; they do not computationally prove the toric moment-map theorem.

Bounded exact-title, exact-question, keyword, and citation searches through 26 August 2026 found the source paper and general literature on *closed* convex hulls, but no later paper deciding Question 7.2 or recording the simple-spectrum theorem above. The result should therefore be treated as a new candidate partial answer pending expert review, not as a claim of certified literature novelty.

References

- [1] X. Mootoo and P. Skoufranis, *Joint majorization in continuous matrix algebras*, J. Operator Theory **92** (2024), 413–438; arXiv:2210.13309.
- [2] C. Lian, *The HHMP decomposition of the permutohedron and degenerations of torus orbits in flag varieties*, arXiv:2309.01747, version 4 (2024).
- [3] F. Sottile, *Toric ideals, real toric varieties, and the algebraic moment map*, Contemp. Math. **334** (2003), 225–248; arXiv:math/0212044.